

# TECHNICAL ARCHITECTURE MANUAL

100% Google Cloud Ecosystem • Cloud Run • Vertex AI • Cloud SQL

ATLAS TRAVEL CLUB LLC • CONFIDENTIAL • PRE-SEED ROUND (\$75K / \$1.75M CAP)

## 1. GOOGLE CLOUD RUN COMPUTE & AUTOSCALING

All platform workloads are containerized using Bun and deployed to Google Cloud Run (us-central1) with fully managed concurrency (80 concurrent requests/instance, min 1, max 50 instances). Ingress is protected by Cloud Load Balancing and Google Cloud Armor edge bot mitigation, enforcing the Rate Parity Compliance Shield by blocking scrapers and applying X-Robots-Tag: noindex at the CDN layer.

## 2. DATABASE & CLOUD SQL AUTH PROXY

The primary database is Google Cloud SQL for PostgreSQL (v16) configured with regional persistent disks and automated 7-day point-in-time recovery. Access is strictly private IP via Cloud SQL Auth Proxy using short-lived IAM-derived OAuth tokens rather than static database passwords, ensuring zero public IP exposure.

## 3. ARTIFICIAL INTELLIGENCE VIA VERTEX AI (GEMINI)

All AI inference is routed natively through Google Cloud Vertex AI using Gemini 1.5 Pro (deep itinerary reasoning, multi-supplier rate aggregation) and Gemini 2.0 Flash (instant conversational turns). The AI Concierge utilizes Vertex AI Function Calling to query RateHawk, Hotelbeds, and Duffel APIs in real-time, keeping live wholesale queries server-side behind authenticated sessions.

## 4. SECRETS & ENVIRONMENT INTEGRATIONS

All credentials (HOTELBEDS\_API\_KEY, RATEHAWK\_KEY\_ID, DUFFEL\_ACCESS\_TOKEN, STRIPE\_SECRET\_KEY, AIRALO\_CLIENT\_SECRET) are stored in Google Secret Manager and mounted directly into the Cloud Run container environment at runtime. Secrets are never checked into version control, never logged, and rotated seamlessly via Secret Manager versioning.